

Analyzing Ellipses: Foci, Axes, and Standard Form

Precalculus / Analytic Geometry Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Find the center, vertices, and foci of an ellipse given key information
- Determine the values of a , b , and c using the relationship $c^2 = a^2 - b^2$ and the major axis length
- Write the standard equation of an ellipse in the form $(x-h)^2/a^2 + (y-k)^2/b^2 = 1$

Problems

1. Two foci of an ellipse are located at $(1, 3)$ and $(7, 3)$. Find the center of the ellipse.

2. The major axis of an ellipse has a length of 10. Find the value of a .

$$2a = 10$$

3. An ellipse has foci at $(0, 2)$ and $(6, 2)$. Find the value of c , which is the distance from the center to each focus.

4. An ellipse has $a = 5$ and $c = 3$. Use the relationship between a , b , and c to find b squared.

$$c^2 = a^2 - b^2$$

5. An ellipse has foci at $(0, 1)$ and $(4, 1)$ and a major axis of length 6. Find the values of a , b , and c .

6. Using the information from Problem 5, write the standard equation of the ellipse with center $(2, 1)$, a squared = 9 (horizontal), and b squared = 5.

$$\frac{(x-2)^2}{9} + \frac{(y-1)^2}{5} = 1$$



7. An ellipse has foci at (2, 0) and (2, 4) and a major axis of length 8. Find the center, c, a, and b squared.

8. Write the standard equation of the ellipse from Problem 7 with center (2, 2), vertical major axis, a squared = 16, and b squared = 12.

$$\frac{(x - 2)^2}{12} + \frac{(y - 2)^2}{16} = 1$$

9. Given the equation of an ellipse, identify the center, the values of a and b, and determine whether the major axis is horizontal or vertical.

$$\frac{(x + 3)^2}{25} + \frac{(y - 4)^2}{9} = 1$$

10. An ellipse has foci at (-1, 2) and (5, 2) and passes through the point (2, 2 + sqrt(5)). Find c, a, b squared, and write the standard equation.



Analyzing Ellipses: Foci, Axes, and Standard Form — Answer Key

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Answer Key

1. Answer: (4, 3)

- The center is the midpoint of the two foci.
- x-coordinate of center: $(1 + 7) / 2 = 4$
- y-coordinate of center: $(3 + 3) / 2 = 3$
- Center = (4, 3)

2. Answer: a = 5

- The relationship between the major axis and a is: $2a = \text{major axis length}$
- $2a = 10$
- $a = 10 / 2 = 5$

3. Answer: c = 3

- First, find the center: midpoint of (0, 2) and (6, 2) = (3, 2)
- The distance from the center (3, 2) to either focus is c.
- $c = |6 - 3| = 3$

4. Answer: $b^2 = 16$

- Use the formula: $c^2 = a^2 - b^2$
- Substitute: $3^2 = 5^2 - b^2$
- $9 = 25 - b^2$
- $b^2 = 25 - 9 = 16$

5. Answer: a = 3, c = 2, $b^2 = 5$

- Center = midpoint of (0,1) and (4,1) = (2, 1)
- c = distance from center to focus = $|4 - 2| = 2$
- $2a = \text{major axis} = 6$, so $a = 3$
- $c^2 = a^2 - b^2 \rightarrow 4 = 9 - b^2 \rightarrow b^2 = 5$

6. Answer: $(x-2)^2/9 + (y-1)^2/5 = 1$

- Standard form: $(x-h)^2/a^2 + (y-k)^2/b^2 = 1$ for a horizontal major axis
- $h = 2$, $k = 1$, $a^2 = 9$, $b^2 = 5$
- Substitute to get: $(x-2)^2/9 + (y-1)^2/5 = 1$

7. Answer: Center = (2, 2), c = 2, a = 4, $b^2 = 12$

- Center = midpoint of (2, 0) and (2, 4) = (2, 2)
- Since foci share the same x-coordinate, the major axis is vertical
- c = distance from center (2, 2) to focus (2, 4) = 2



- $2a = 8$, so $a = 4$
- $c^2 = a^2 - b^2 \rightarrow 4 = 16 - b^2 \rightarrow b^2 = 12$

8. Answer: $(x-2)^2/12 + (y-2)^2/16 = 1$

- For a vertical major axis, the larger denominator (a^2) goes under the y term
- $h = 2$, $k = 2$, $a^2 = 16$ (under y), $b^2 = 12$ (under x)
- Standard form: $(x-2)^2/12 + (y-2)^2/16 = 1$

9. Answer: Center $(-3, 4)$, $a = 5$, $b = 3$, horizontal major axis

- Rewrite: $(x - (-3))^2/25 + (y - 4)^2/9 = 1$
- Center = $(h, k) = (-3, 4)$
- $a^2 = 25 \rightarrow a = 5$; $b^2 = 9 \rightarrow b = 3$
- Since $25 > 9$ and 25 is under the x term, the major axis is horizontal

10. Answer: $c = 3$, $a^2 = b^2 + 9$; $b^2 = 5$, $a^2 = 14$; equation: $(x-2)^2/14 + (y-2)^2/5 = 1$

- Center = midpoint of $(-1, 2)$ and $(5, 2) = (2, 2)$
- c = distance from center to focus = $|5 - 2| = 3$
- The point $(2, 2 + \sqrt{5})$ is on the ellipse; substitute into standard form: $(2-2)^2/a^2 + (2+\sqrt{5}-2)^2/b^2 = 1$
- This gives: $0 + 5/b^2 = 1$, so $b^2 = 5$
- Use $c^2 = a^2 - b^2$: $9 = a^2 - 5$, so $a^2 = 14$
- Standard equation: $(x-2)^2/14 + (y-2)^2/5 = 1$

